TSM 099 Arithmetic Test

Practice 3 | Version 202610122020.18

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right.

QUESTIONS	ANSWERS
1. Simplify the square root. $\sqrt{20}$	1. Radical 20 pts
2. Substitute a = 2, b = -5, c = 2 into the formula. Evaluate both branches. $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	2. Both values 20 pts
3. Evaluate each side separately at x = 8, -8, 0. Record the left value, right value, and whether they are equal. State what the tested values establish; do not solve the equation. $\frac{x^2 - 16}{x - 4} = \frac{x^2 + 10x + 25}{x + 5}$	3. x = 8: left / right / equal? 4 pts x = -8: left / right / equal? 4 pts x = 0: left / right / equal? 4 pts Conclusion 8 pts
4. Reduce each fraction to lowest terms. (a) $\frac{21}{35}$ (b) $\frac{70}{110}$	4. (a) 10 pts (b) 10 pts
5. Evaluate the principal square root in each part. (a) $\sqrt{49}$ (b) $\sqrt{-81}$	5. (a) 10 pts (b) 10 pts

Instructor answer key: TSM 099 Arithmetic Test

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Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

TEST PAGE 1

1. Radical

$2\sqrt{5}$

20 pts

2. Both values

$x = \frac{1}{2},\ 2$

20 pts

3. x = 8: left / right / equal?

Left 12; right 13; no — 4 pts

x = -8: left / right / equal?

Left -4; right -3; no — 4 pts

x = 0: left / right / equal?

Left 4; right 5; no — 4 pts

Conclusion

None of these values works; this alone does not prove no solution. — 8 pts

4. (a)

$\frac{3}{5}$

10 pts

(b)

$\frac{7}{11}$

10 pts

5. (a)

7

10 pts

(b)

9i

10 pts